

NAME

`sync_operator_field.py` – correct the operator field inside `.set` files to match the schedlist owner

SYNOPSIS

`sync_operator_field.py --data-dir DIR --quarantine-dir DIR [--execute | --restore RUN_DIR]`

DESCRIPTION

`sync_operator_field.py` finds `<id>.set` files whose internal **operator** field disagrees with the **owner** recorded for that id in *schedlist*, and rewrites the `.set` file in place to match.

Each `<id>.set` file contains a pipe-delimited line beginning **operator|**, whose second-to-last field holds the Symphony operator id that field is meant to agree with the owning template's **owner** field in *schedlist* (field index 6). Drift between the two can accumulate for reasons outside this tool's scope; this tool detects and fixes it, one field, one file, at a time.

This is field-level surgery, not a quarantine-and-remove operation like `remove_orphans.py(1)` or `remove_stale_templates.py(1)`: no files are moved out of `--data-dir`. `--quarantine-dir` is used only to hold the run's *manifest.csv*, which records enough to reverse every field edit.

OPTIONS

`--data-dir DIR`

Path to the *rptsched* data directory to scan and modify. Required in every mode, including `--restore`.

`--quarantine-dir DIR`

Path to the directory that holds run subfolders (and their *manifest.csv* files). Created if it does not already exist. Required in every mode.

`--execute`

Perform the field rewrites. Without this flag the script runs in dry-run mode and writes nothing.

`--restore RUN_DIR`

Reverse a previous `--execute` run's field edits using `RUN_DIR/manifest.csv`. Mutually exclusive with `--execute`.

MODES**Dry run (default)**

For every id present in *schedlist*, the script looks for a matching `<id>.set` file and extracts its **operator** field:

- If the `.set` file is missing, or has no **operator|** line, the id is skipped with a **WARNING** on stderr (reason given) and excluded from the mismatch count entirely – it is not something this tool can fix.
- Otherwise, if the extracted operator differs from the *schedlist* owner, it is reported as a mismatch: **id: old_operator -> new_operator**.

Nothing on disk is touched in dry-run mode.

Execute

With `--execute`, the script re-finds mismatches (skips are reported the same way as dry run) and, if any exist, creates `--quarantine-dir/operators_TIMESTAMP`, then for each mismatched id in id order:

1. Rewrites the `.set` file's **operator|** line in place – via a temp file plus atomic rename, preserving the original file's permissions and mtime metadata – setting the operator field to the *schedlist* owner value.
2. Appends a row to the run's *manifest.csv* (id, old_operator, new_operator).

If zero mismatches are found, the script prints a message and exits 0 without creating a run subfolder. If a rewrite fails partway through, the ids already corrected keep their manifest row (so they're all individually restorable), a partial *manifest.csv* is written, and the script exits non-zero.

Restore

With **--restore** *RUN_DIR*, the script reads *RUN_DIR*/manifest.csv and, for each row, reads the id's **current** operator value and compares it against the row's recorded **old_operator** and **new_operator**:

- Current equals **old_operator** — already restored (by an earlier attempt, or never actually changed); skipped.
- Current equals **new_operator** — this is the normal case; the field is rewritten back to **old_operator** and counted as restored.
- Current matches neither — something else modified the field since this tool last touched it. The script prints an error naming the id and all three values, and aborts immediately with a non-zero exit rather than guessing which value is correct.

This makes restore idempotent and resumable in the same spirit as **remove_orphans.py**(1)'s file-restore: safe to re-run on the same *RUN_DIR*, and it stops rather than overwrites when something looks wrong.

EXIT STATUS

- 0** Success – including a clean dry run, an execute run that found zero mismatches, or a restore that skipped everything because it was already complete.
- 1** A manifest could not be read (wrong file / wrong tool's run directory), a field rewrite failed, or a restore hit an id whose current value matched neither the recorded old nor new value.

FILES

<data-dir>/schedlist

Read only – supplies the **owner** value each **.set** file's **operator** field is compared and synced against. Never modified by this tool.

<data-dir>/<id>.set

Rewritten in place (atomically) for each mismatched id, both by **--execute** (old operator -> new) and by **--restore** (new operator -> old, when applicable).

<quarantine-dir>/operators_<timestamp>/manifest.csv

Written by **--execute**; read by **--restore**. Columns: id, old_operator, new_operator. Distinct schema from the **remove_orphans.py**(1) / **remove_stale_templates.py**(1) manifest – the two are not interchangeable, and **--restore** rejects a manifest with the wrong columns rather than misinterpreting it.

EXAMPLES

Preview operator/owner mismatches:

```
sync_operator_field.py --data-dir ./rptsched --quarantine-dir ./quarantine
```

Apply the corrections:

```
sync_operator_field.py --data-dir ./rptsched --quarantine-dir ./quarantine --execute
```

Undo a specific run:

```
sync_operator_field.py --data-dir ./rptsched --quarantine-dir ./quarantine --restore ./quarantine/operators_20260826_140501
```

SEE ALSO

remove_orphans.py(1), **remove_stale_templates.py**(1)

CAVEATS

Requires Python 3.6.8, standard library only.

Unlike the other two tools, nothing is ever moved out of **--data-dir** – there is no quarantined copy of a **.set** file to fall back on, only the manifest's record of the old and new field values. If a **.set** file is edited by something else between **--execute** and **--restore**, restore for that id aborts rather than clobbering the unexpected value, leaving it for the operator to check that id and resolve the mismatch by hand.